BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions

Note: Q. 1 Choose the correct answer for each from the given options:

Q1. If A superset B then AUB= _____

(a) A
(b) B
(c) Ø
(d) U

Q2. Thp mean proportional between a and b is

(a) root(ab)
(b) ab
(c) a/b
(d) -ab

Q3. For matrix A, (A^t)^-1 _____

(a) A^-2
(b) A
(c) A^-1
(d) A^2

Q4. If α,β are the roots of x^2 - 2x - 15 = 0, then the value α^2 + β^2 = _____

(a) 34
(b) -34
(c) 26
(d) -26

Q5. An improper fraction can be reduced into proper fraction by

(a) Addition
(b) Multiplication
(c) Subtraction
(d) Division

Q6. The ungrouped data must be ordered first to find _____

(a) A.M
(b) Mode
(c) Median
(d) Range

Q7. if 5 cm and 12 cm are two sides of a right angled triangle. Then hypotenuse is:

(a) 16 cm
(b) 15 cm
(c) 14 cm
(d) 13 cm

Q8. Two equilateral triangles are also_____

(a) Congruent
(b) Similar
(c) Proportional
(d) Equivalent

Q9. Inscribed angle of _____ is obtuse.

(a) Minor arc
(b) Major arc
(c) Semi-circle
(d) All of these

Q10. Angle between tangent and radial segment at the point of contact angle.

(a) Right
(b) Obtuse
(c) Acute
(d) Reflex

Q11. The sum fo central angles of all the arcs of a circle is _____.

(a) 90 degrees
(b) 108 degrees
(c) 120 degrees
(d) 135 degrees

Q12. The ungrouped data must be ordered first to find

(a) A.M
(b) Mode
(c) Median
(d) Range

Q13. If 5 cm and 12 cm are two sides of a right angled triangle. Then hypotenuse is:

(a) 16 cm
(b) 15 cm
(c) 14 cm
(d) 13 cm

Q14. Two equilateral triangles are also

(a) Congruent
(b) Similar
(c) Proportional
(d) Equivalent

Q15. Inscribed angle of is obtuse.

(a) Minor arc
(b) Major arc
(c) Semi-circle
(d) All of these

Q16. Angle between tangent and radial segment at the point of contact angle.

(a) Right
(b) Obtuse
(c) Acute
(d) Reflex

Q17. The maximum number of common tangents between two circles touching externally is _____.

(a) 0
(b) 1
(c) 2
(d) 3

Q18. Diameter divides the circle into _____ parts.

(a) Two
(b) Three
(c) Four
(d) All of these

Q19. Each interior angle of regular hexagon is equal to _____

(a) 90 degrees
(b) 108 degrees
(c) 120 degrees
(d) 135 degrees

Section-B

> Note: Solve any EIGHT of the following questions. Each question carries 04 marks

Q2. Verify the commutative property of union for the set: A= {a, b, c, d, e} and B={a, e, b, d}

Q3. Find the value of "a" if the ratios 3a + 4 : 2a +5 and 4:3 are equal,

Q4. Prove that a:b=c:d if a^2 - b^2: a^2 + b^2 = ac bd : ac + bd

Q5. For what values of p and q the roots of quadratic equation (2p - 4)x - (3q + 5) = 0

Q6. Solve the systems of equation, 2xy = 3 and x y^2 = 2

Q7. Show that the diameters of a circle bisect each other.

Q8. Find the values of x and y if (x - 5,10) = (11, -7)

Q9. For what value of x, the matrix [[5 - x, x + 1],[2, 4]] is singular?

Q10. Find all the cube roots of 64.

Q11. If A={1,2}, B={2,3} and C={1,3,5}, then find (i) A x B (ii) A x (BUC)

Q12. Find the value of m, if the roots of the equation x^2 - 5x + 2m = 0 differ by 1.

Q13. How far from centre of the radius 3 cm, a tangent segment of length 10 cm can be drawn?

SECTION-C

> Note: NOTE: Solve any FOUR of the following questions. Each question carries 07 marks.

Q14. Verify De Morgan's Laws using Venn diagram if A = {1,3,5,7,9}, B= {5,6,7,8} and U={1, 2, 3,.....10}

Q15. Find the solution by matrix inversion method rule 2x + y = 14; -4x + y = 28

Q16. Show that (1 - ω) (1 + ω^2) (1 + ω^4) (1 + ω^8) = (ω + ω^2)^4

Q17. The length of sides of triangle, a=16, b=30 and c=34, verify that the triangle is right triangle.

Q18. Resolve (6x - 5)/((x^2 + 10)(x +1)) into partial fractions.

Q19. Prove that, if a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point.